NeuroForge: A Self-Correcting, Geometry-Native Neural CFD Engine with Calibrated Physics-Residual Trust

Ali Jabbary
Independent Researcher
author@example.com

Abstract

Machine-learning surrogates for computational fluid dynamics (CFD) predict steady flow fields over engineering geometries orders of magnitude faster than classical solvers, but they emit a single field with no built-in way for a user to know whether to trust it—especially out of distribution, where early-design exploration lives. A natural idea is to close the loop with the governing physics: compute the discretised steady-RANS residual of the prediction, use it both to *flag* unreliable regions and to *drive a correction* back toward a valid solution. We implement this full pipeline in **NeuroForge CFD**, an open-source, CPU-first package, and stress-test it on real AirfRANS over three seeds. Our central finding is a clean dissociation: **the physics residual is an excellent trust signal but a poor correction objective.** As a *detector*, the residual is a calibrated proxy for error—its rank correlation with field error rises from 0.40 to 0.83 under our corrector in-distribution, and, critically, it stays informative under distribution shift (angle-of-attack split: the bare backbone's residual–error correlation collapses to 0.31 while the corrected model holds it at 0.75). As a *fixer*, the residual fails three independent ways: a contractive deep-equilibrium (DEQ) corrector is flat on volume velocity (`mse_u` 3.46 vs 3.48) and *worse* on cross-stream velocity (`mse_v` 0.88 vs 0.39); running the loop for more iterations *raises* the PDE residual while *lowering* field error; and a backtracking acceptance test that only admits residual-reducing steps accepts essentially none on a trained corrector. Minimising the residual diverges from minimising error. Building on the detector result, we package a **distribution-free conformal trust layer** that attains per-channel coverage of 0.91/0.93/0.94 at the 0.90 target in-distribution (a guarantee that holds only under exchangeability) and that empirically retains target coverage out-of-distribution because the underlying uncertainty inflates appropriately under shift—an empirical observation, not a guarantee. The trust layer is backbone-agnostic. For honest context we report a fair, matched-budget baseline: our grid backbone trails a SOTA point-cloud transformer (Transolver) by roughly 4–60× across volume and surface MSE channels. We make **no competitiveness claim**; the contribution is the residual-as-trust-signal phenomenon and the calibrated trust layer, and applying them to a SOTA backbone is explicit future work.

Keywords: neural operators, surrogate CFD, physics residuals, uncertainty quantification, conformal prediction, trust calibration, airfoil aerodynamics.

1 Introduction

Classical CFD resolves the governing PDEs by iterating a discretised system until its residuals fall below a tolerance. It is accurate and trusted, but slow: meshing and a converged steady-RANS solve over a new geometry can take minutes to hours, throttling the rapid design-space exploration that early engineering most needs. Neural surrogates promise the opposite trade—millisecond inference—and a vigorous literature now maps geometry plus boundary conditions directly to flow fields with neural operators and attention models.

The problem is **trust**. A trained surrogate is a one-shot function: it emits a field with no self-assessment, no convergence signal, and no recovery path when it extrapolates. On a geometry far from the training set—the common case in design—the user cannot distinguish a faithful prediction from a confident hallucination.

The governing physics offers an obvious lever. The steady incompressible RANS residual of any candidate field is computable directly from the field, with no ground truth. This invites a tempting two-for-one: use the residual *both* to flag where the surrogate is wrong (a detector) *and* to correct it by driving the residual down (a fixer). The fixer half is the premise behind learned solver-correctors and residual-conditioned refinement. This paper asks, and answers empirically, whether the residual can play both roles.

Research question. *Is the steady-RANS physics residual of a neural-CFD prediction a reliable signal of where the prediction is wrong, and can the same residual be used as an objective to make the prediction right?*

Thesis (one sentence). ***The physics residual is an excellent trust signal but a poor correction objective:*** *it is a calibrated, distribution-shift-robust detector of error, yet minimising it diverges from minimising error.*

We arrive at this thesis the hard way. We built the full predict→verify→estimate-uncertainty→correct→fallback engine, pre-registered hypotheses, and ran a 3-seed ablation on real AirfRANS plus an out-of-distribution study and two formal certificates. The engine's *correction* machinery is the part that disappoints, and we report that as a finding rather than hide it; the engine's *detection* and *calibration* machinery is the part that survives, and it survives under distribution shift, which is exactly where it is needed.

1.1 Contributions

In priority order:

1. **Residuals do not fix (headline, a negative result).** The physics residual is a poor correction objective, shown three independent ways. (i) A contractive DEQ corrector is flat on volume `mse_u` (3.46 vs 3.48) and **worse** on `mse_v` (0.88 vs 0.39) and `mse_p` (3833 vs 2445) than its own backbone, and a feed-forward corrector helps on nothing (subsection 5.2, Table 1). (ii) Sweeping the number of correction iterations *raises* the PDE residual (0.11 → 0.62) while *lowering* field error (`mse_u` 3.92 → 2.29)—the residual and the error move in opposite directions (subsection 5.4, Figure 4). (iii) A backtracking acceptance test that only admits residual-reducing steps accepts essentially zero steps on a trained corrector (subsection 5.4). Minimising the residual diverges from minimising error.

2. **Residuals detect (the solid, useful core).** The same residual is a strong, calibrated error signal. Its rank (Spearman) correlation with field error rises from 0.40 to 0.83 under the DEQ corrector in-distribution (subsection 5.2), and—the single strongest result—it **holds under distribution shift**: on the angle-of-attack split the bare backbone's residual–error correlation collapses to 0.31 while the corrected model keeps it *regime-invariant* at 0.75 (subsection 5.3, Figure 2). The trust signal stays reliable precisely in the regime where the one-shot model loses it.

3. **A calibrated, backbone-agnostic conformal trust layer.** Split-conformal calibration of the predictive uncertainty attains per-channel coverage of 0.91/0.93/0.94 against a 0.90 target in-distribution (a distribution-free guarantee under exchangeability), and *empirically* retains target coverage out-of-distribution because the MC-dropout uncertainty inflates under shift—an empirical observation, **not** a guarantee where exchangeability fails (subsection 5.5, Figure 5–6). The layer depends only on a model's predictions and uncertainty, so it is backbone-agnostic.

4. **An honest, matched-budget benchmark.** We report a fair comparison against a SOTA point-cloud transformer (Transolver) trained on the same data with the same budget and scored identically

(subsection 5.6, Table 4). Our grid backbone trails it by roughly 4–60× across volume and surface MSE channels. We include this as context and make **no competitiveness claim**; the SOTA-backbone version of the trust layer is future work.

5. **A reproducible, CPU-first open-source package** (`neuroforge-cfd`) with a frozen I/O contract, interchangeable backbones, a zero-download synthetic potential-flow data generator, an AirfRANS loader, the ablation/certificate harness, and the figures—so every claim runs on a laptop and every certificate is reproducible from a committed script.

We are explicit throughout about what is **measured** versus **assumed**, retain the retraction of stale single-seed numbers from earlier drafts (subsection 5.2), and make no competitiveness or out-of-distribution-guarantee claim anywhere.

2 Related Work

Neural operators (FNO / Geo-FNO). Fourier Neural Operators learn a resolution-agnostic mapping between function spaces by parameterising a global convolution in the spectral domain (Li et al., 2021). Geo-FNO (Li et al., 2022) extends this to irregular geometries via a learned deformation to a latent uniform grid where the FFT is valid. These are strong *one-shot* operators with no self-assessment; NeuroForge implements both as backbones and adds the trust layer around them.

Point-cloud and physics-attention operators (DoMINO, Transolver). DoMINO (Ranade et al., 2025) is a decomposable multi-scale point-cloud operator for external aerodynamics on large industrial meshes. Transolver (Wu et al., 2024) replaces quadratic attention with attention over learnable *physics slices*, giving linear cost and strong accuracy on PDE benchmarks; Transolver++ (Luo et al., 2025) scales it to million-scale meshes. We use a matched-budget Transolver as our honest baseline (subsection 5.6); the trust-signal phenomenon we characterise is independent of the backbone, and putting it on such a model is future work.

Learned solver-correctors and the "fixer" premise. Several lines of work use the discretised residual or a coarse-solution error as a *correction objective*: learned PDE solvers with convergence guarantees (Hsieh et al., 2019), deep-equilibrium operators for steady PDEs (Marwah et al., 2023), iterative refiners (Lippe et al., 2023), and learned residual-error correctors (Anonymous, 2023). PINNs (Raissi et al., 2019) embed the residual in the training loss. NeuroForge implements a contractive DEQ corrector and a feed-forward residual-conditioned corrector squarely in this family. **Our contribution to this line is a negative result:** on a real RANS benchmark, residual minimisation does not track error minimisation—the corrector trades volume accuracy for a stronger residual signal—which sharpens *when* the residual-as-objective premise holds.

Uncertainty quantification and conformal prediction for PDE surrogates. Deep ensembles (Lakshminarayanan et al., 2017) and MC-dropout (Gal & Ghahramani, 2016) are standard epistemic estimators; conformal prediction gives distribution-free coverage and has been applied to operator learning (Ma et al., 2024). We do **not** propose a new UQ method. Our trust layer applies split-conformal calibration to MC-dropout uncertainty; the new content is the *characterisation* of when this holds for CFD surrogates—in particular that coverage survives out-of-distribution empirically via uncertainty inflation, while the formal guarantee does not transfer once exchangeability fails.

Benchmarks (AirfRANS). AirfRANS (Bonnet et al., 2022) provides $\sim$ 1000 incompressible steady-RANS simulations over NACA airfoils with `full`, `scarce`, `reynolds`, and `aoa` splits designed to probe generalisation. We use `full` for in-distribution evaluation and the regime-disjoint `reynolds`/`aoa` splits for the out-of-distribution study.

2.1 Positioning

The closest prior art treats the physics residual primarily as a *correction objective* (PINNs, FNO-DEQ, learned correctors) or treats UQ in isolation from physics. NeuroForge's distinctive question is whether the residual is better used as a *detector* than as a *fixer*, evaluated head-to-head on the same model and data. Our answer—detector yes, fixer no—and the resulting calibrated trust layer are the contribution. The engine's no-harm/contraction machinery (section 3) is retained because it is correct and reproducible, but it is presented as a mechanism we analyse, not as the headline claim.

3 Method

3.1 Pipeline overview

The engine wraps a one-shot backbone in a verify–estimate–correct–fall-back loop. We describe all components for completeness; section 5 shows empirically that the *detection* and *calibration* components carry the contribution while the *correction* component is limited.

```
CAD / STL / airfoil + BCs
        |
        v  geometry-native encoding (SDF + solid mask + coords + freestream + log Re)
   x in R^{7 x H x W}
        |
        v  neural-operator / transformer backbone  f_theta
   yhat in R^{4 x H x W}  (u, v, p, nu_t)        <---------------+
        |                                                        |
        v  physics residual checker  R(.)                        |  correction
   continuity, momentum_x, momentum_y, BC violation              |  loop
        |                                                        |  (analysed, limited)
        v  uncertainty  sigma(.)  (MC-dropout / deep ensemble)   |
        |                                                        |
        v  conformal trust layer  +  trust map T                 |
        |   (the surviving contribution)                         |
        v  correction operator  c_phi -> Delta ------------------+
        |   (DEQ fixed point, or feed-forward under acceptance test)
        v  (only if a region stays low-trust)
   uncertainty-gated classical CFD patch  (interface)
        |
        v  flow field . Cp . forces . uncertainty/residual/trust maps . history
```

3.2 Geometry-native encoding and governing equations

A `FlowCase` (geometry + boundary conditions + fluid + domain) is encoded into a fixed channel-first stack $x \in \mathbb{R}^{7\times H\times W}$ in the frozen `INPUT_CHANNELS` order $(\texttt{sdf}, \texttt{mask}, x, y, u_{\text{in}}, v_{\text{in}}, \log \text{Re})$: the signed distance to the body surface (negative inside the solid), a fluid/solid mask, normalised cell coordinates, the freestream velocity broadcast over the grid, and $\log_{10}\text{Re}$. The backbone outputs `OUTPUT_CHANNELS` (u, v, p, ν_t), where p is the **kinematic** pressure p/ρ.

The verifier evaluates the steady, incompressible, 2-D RANS equations in primitive form on the **physical (denormalised)** fields, with effective viscosity $\nu_{\text{eff}} = \nu + \nu_t$, pointwise:

$$\text{continuity:} \quad r_c = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}, \tag{1}$$

$$x\text{-momentum:} \quad r_x = u\,\frac{\partial u}{\partial x} + v\,\frac{\partial u}{\partial y} + \frac{\partial p}{\partial x} - \nu_{\text{eff}}\,\nabla^2 u, \tag{2}$$

$$y\text{-momentum:} \quad r_y = u\,\frac{\partial v}{\partial x} + v\,\frac{\partial v}{\partial y} + \frac{\partial p}{\partial y} - \nu_{\text{eff}}\,\nabla^2 v. \tag{3}$$

Because p is kinematic, no explicit density appears. Derivatives use central finite differences with one-sided stencils at the borders; residuals are zeroed inside the solid. A boundary-condition violation map r_{bc} adds a **no-slip** term penalising velocity magnitude in the thin fluid band adjacent to the wall (weighted by

$\exp(-|\mathrm{sdf}|/\ell)$, $\ell \approx 3$ cells) and a **far-field** term penalising deviation from (u_∞, v_∞) on the outer border ring. We stress the standard caveat that drives our whole study: a low residual is **necessary but not sufficient** for correctness—a smooth near-freestream field can have near-zero residual yet be entirely wrong—so the residual is a *consistency monitor*, and whether it tracks error is the empirical question section 5 answers.

3.3 Trust map and conformal calibration (the surviving contribution)

Raw trust map. The combined PDE-residual magnitude $\rho_{\mathrm{res}} = \sqrt{r_c^2 + r_x^2 + r_y^2 + r_{bc}^2}$ and a per-cell predictive uncertainty σ are each mapped to $[0, 1]$ (using an absolute physical reference scale $s = U_\infty^2/L + U_\infty/L$ when available, else a robust 95th-percentile normalisation with an absolute floor) and fused,

$$e = \mathrm{clip}\big(w_r\,\hat{r} + w_u\,\hat{\sigma},\ 0,\ 1\big), \qquad T = 1 - e, \tag{4}$$

with $w_r = 0.6$, $w_u = 0.4$, giving a traffic-light field (green $e < 0.15$, red $e > 0.45$, else yellow). Uncertainty σ is the channel-mean standard deviation from a deep ensemble or MC-dropout.

Split-conformal calibration. Raw σ is uncalibrated, so a threshold on it carries no guarantee. We add split-conformal calibration (cf. Ma et al., 2024): on a held-out calibration set we compute per-channel nonconformity scores $s = |\hat{y} - y|/\sigma$ and take the finite-sample-corrected $(1 - \alpha)$ quantile q; the band $q \cdot \sigma$ then satisfies the distribution-free coverage guarantee in Proposition 1.

Proposition 1 (Distribution-free coverage under exchangeability)**.** *If the calibration scores and a test score are exchangeable, then the split-conformal band $q \cdot \sigma$ satisfies $\mathbb{P}\big(|\hat{y} - y| \le q\,\sigma\big) \ge 1 - \alpha$.*

This converts the trust threshold from a hand-picked constant into a band with a known in-distribution coverage level. subsection 5.5 reports the measured coverage, the coverage-vs-α sweep, and the out-of-distribution behaviour, with the exchangeability caveat made explicit.

3.4 The correction loop (analysed, shown to be limited)

The engine offers two correction operators and a backtracking acceptance test. subsection 5.2 and subsection 5.4 show empirically that none of them turns the residual into a useful correction objective; we describe them precisely so the negative result is interpretable.

Feed-forward corrector + backtracking acceptance test. A small residual CNN c_ϕ predicts an additive correction conditioned on the current field, its current 3-channel residual map, and the geometry: $\Delta_k = c_\phi(y_k, R(y_k), x)$. A candidate $y_{k+1} = y_k + s\,\Delta_k$ is accepted only if it does not increase the residual norm $N(y) = \sqrt{r_c^2 + r_x^2 + r_y^2}$:

$$N(y_k + s\,\Delta_k) \le N(y_k) + \varepsilon, \tag{5}$$

with s halved up to four times on failure. By construction $N(y_0) \ge N(y_1) \ge \cdots \ge N(y_K)$—the residual norm is monotone non-increasing across accepted steps. This is the convergence discipline of a classical solver. Its weakness is exactly the one we exploit as a finding: the test is satisfiable by the identity (zero step), so a corrector that cannot reduce the residual is simply rejected. Empirically (subsection 5.4) a trained corrector accepts essentially no steps, because reducing the field error and reducing the PDE residual are not the same objective.

Contractive DEQ corrector. To give the loop a genuine convergence guarantee in its *own* iteration variable, the correction δ is defined as the fixed point of a learned operator

$$\delta^* = T_\theta(\delta^*; c), \quad T_\theta(\delta; c) = \kappa \cdot g_\theta([\delta, c]), \quad c = [\hat{y}, r(\hat{y}), \mathrm{geom}], \tag{6}$$

where g_θ is a CNN whose layers are spectrally normalised (each $\le$ 1-Lipschitz) and $\kappa < 1$.

Proposition 2 (Banach contraction of the corrector)**.** *With g_θ each-layer 1-Lipschitz and $\kappa < 1$, T_θ is a κ-contraction in δ. By the Banach fixed-point theorem the equilibrium δ^* exists, is unique, and the iteration converges geometrically, $\|\delta_k - \delta^*\| \le \kappa^k \|\delta_0 - \delta^*\|$.*

The construction underlying Proposition 2 is a Deep-Equilibrium model (Bai et al., 2019) with the Lipschitz constant controlled by spectral normalisation (Winston & Kolter, 2020), trained with Jacobian-Free Backpropagation (Fung et al., 2022). Crucially, g_θ is trained on **data** (it targets the correction toward truth, with the residual as an input *feature*) rather than by minimising the residual; at inference the converged δ^* is applied directly, **without** the backtracking acceptance test. We verify this contraction empirically (subsection 5.5: measured factor $0.78 < \kappa = 0.9$). Two scopes must be kept distinct, and subsection 5.4 leans on the distinction: the DEQ contraction is in the *correction variable* δ, whereas the *PDE residual of the output field* is a different quantity that, as we show, can rise even as δ converges and the field error falls.

The feed-forward corrector is trained (backbone frozen) so that $c_\phi(\hat{y}_{\text{norm}}, R_{\text{norm}}, x) \approx y^\star_{\text{norm}} - \hat{y}_{\text{norm}}$, with the final convolution initialised near zero for a stable first iteration.

3.5 Uncertainty-gated classical fallback

After the loop, if the maximum uncertainty exceeds a threshold, the low-trust fluid region is handed to a `ClassicalFallback`. In the present release this is an **interface with a `stub` backend** that reports what would run (and the region size) without invoking an external solver; the `openfoam`/`su2` backends raise `NotImplementedError` with setup guidance. This keeps the engine importable and runnable with nothing installed while fixing the integration seam; it is not part of any quantitative claim.

4 Implementation

NeuroForge is an open-source, **CPU-first**, pure-Python package (`neuroforge-cfd`, Python $\geq$ 3.10, NumPy/SciPy/PyTorch/Matplotlib). Importing the package caps BLAS/OMP/MKL thread counts to one by default to avoid catastrophic oversubscription on low-core machines, and does no heavy work. The codebase is organised around a **frozen I/O contract** in `core/`.

Module map. `core/` holds the frozen data contracts (`FlowCase`, `FlowField`, `Diagnostics`, `SolveResult`) and `Config`. `geometry/` builds the 7-channel input (NACA generation, SDF/mask rasterisation, `encode_case`). `data/` has the synthetic generator, the AirfRANS loader, the rasteriser, and the `datamodule` (`Normalizer`/loaders). `models/` provides `FNO2d`, `GeoFNO`, a Transolver-style `PhysicsTransformer`, `UNet`/`DeepONet` baselines, `LocalCorrectionNet`, the `DEQCorrector`, and the `DeepEnsemble`/`MCDropoutUQ` wrappers, all in a string-keyed registry. `physics/` has the differential operators, the `PhysicsChecker`, `trust_map`, force/Cp/error metrics, the conformal calibration, and the differentiable `physics_residual_torch`. `solver/` has `Predictor`, `NeuroForgeEngine`, `neural_residual_iteration`, and `ClassicalFallback`. `train/`, `viz/`, `cli.py`, `app/` provide training, plotting/report, the CLI, and a Streamlit UI.

Fixed I/O contract. Inputs are always the 7 channels ($\texttt{sdf}, \texttt{mask}, x, y, u_{\text{in}}, v_{\text{in}}, \log \text{Re}$); outputs always the 4 channels (u, v, p, ν_t). Fields are (`ny`,`nx`) `float32`; network tensors channel-first ($B, C, \texttt{ny}, \texttt{nx}$). Pressure is kinematic; residuals are computed on denormalised fields with $\nu_{\text{eff}} = \nu + \nu_t$. Fixing this contract is what makes the backbones interchangeable and the trust layer backbone-agnostic.

Backbones. `FNO2d` is a faithful spectral FNO; `GeoFNO` adds a geometry-gated conditioning of the lifted features; `PhysicsTransformer` implements Transolver-style physics-slice attention (linear in grid points); `UNet`/`DeepONet` are baselines.

Training loss. The `CompositeLoss` sums a masked data MSE over fluid cells, a differentiable physics-residual term on the denormalised fields, and a no-slip BC term. The corrector is trained separately with the backbone frozen (subsection 3.4).

Synthetic potential-flow generator (zero-download reproducibility). `SyntheticRANS` superposes a Hess–Smith source-panel potential core (with a Kutta-enforcing bound vortex), an algebraic near-wall no-slip ramp and Gaussian wake deficit, kinematic Bernoulli pressure, and a mixing-length ν_t, with desingularised

Table 1: In-distribution AirfRANS `full` ablation, 3 seeds, mean ± std (population std). Lower MSE is better; ρ closer to 1 is better; resid↔err $\rho > 0$ supports the trust signal. Bold marks the best arm per column among the physics-loss arms; the physics-loss-free arm (starred where best overall) is reported separately because it removes the physics objective entirely. See Figure 1.

arm	`mse_u`	`mse_v`	`mse_p`	surf. `mse_p`	ρ_{C_l}	ρ_{C_d}	resid↔err ρ
backbone	3.479±0.105	**0.385±0.012**	**2444.8±120.7**	548823±27946	0.9868±0.0005	0.895±0.013	0.397±0.003
backbone (no phys. loss)	**1.995±0.042**	0.323±0.008*	1963.5±55.5*	1123649±104822	**0.9920±0.0002**	**0.945±0.008**	0.605±0.016
backbone + local corr.	3.482±0.058	0.398±0.007	2469.4±71.0	541534±20453	0.9854±0.0017	0.888±0.012	0.389±0.031
backbone + DEQ corr.	3.457±0.811	0.880±0.064	3832.7±331.3	**361681±12273**	0.9856±0.0017	0.923±0.014	**0.827±0.002**

kernels and light smoothing. The fields are physically plausible and continuity-respecting but **analytic**—a smoke-test/reproducibility substrate, not solver ground truth. **No quantitative claim in this paper rests on synthetic data**; all reported numbers (section 5) are on real AirfRANS.

AirfRANS loader. `load_airfrans` reads each simulation's point cloud, reconstructs the airfoil loop, rasterises the targets onto a structured crop, and returns (`FlowCase`, `FlowField`) pairs for all four splits.

5 Experiments

All quantitative results are on **real AirfRANS**. Unless noted, the in-distribution and out-of-distribution ablations use the pre-registered protocol (`benchmarks/ablation.py`, `docs/EXPERIMENTS.md`) over **3 seeds (0, 1, 2)**, reported as mean ± std (population std, `ddof=0`; sample std at $n = 3$ widens bars by $\approx 1.22\times$). Metrics follow the AirfRANS community protocol (`evaluate_cases`): per-channel volume MSE (lower is better, well-conditioned), surface-pressure MSE on the body, Spearman rank correlation of the force coefficients ρ_{C_l}, ρ_{C_d} (closer to 1 is better—what early-design ranking needs), and `residual_error_spearman` (> 0 means a low residual tracks low error). The ν_t channel is near-degenerate at this scale (MSE $\approx 5 \times 10^{-8}$ for every arm) and is omitted from the tables (see section 6).

With $n = 3$ we report **per-seed sign-consistency** plus effect-size magnitude rather than p-values, which are meaningless at $n = 3$. An effect called "robust" below is sign-consistent across all 3 seeds with a large effect size and (where stated) non-overlapping per-seed distributions.

5.1 Setup

The backbone is an FNO trained on AirfRANS `full` (800 train / 200 test, 80 epochs). Four arms: `backbone`, `backbone (no physics loss)`, `backbone + local corrector` (feed-forward, with the acceptance test), and `backbone + DEQ corrector`. The certificate runs (subsection 5.5) use a dropout-enabled FNO (width 48, 4 layers, 20 modes, dropout 0.05) with a DEQ corrector ($\kappa = 0.9$, damping 0.5), trained $40 + 15$ epochs at resolution 128.

5.2 Residuals do not fix, part 1: the in-distribution ablation

The fixer fails on accuracy. The DEQ corrector is **flat** on volume `mse_u` (3.457 vs backbone 3.479; the per-seed deltas are sign-inconsistent, 2/3, so this is noise, not an improvement) and **robustly worse** on `mse_v` (0.385 → 0.880, +129%, 3/3, non-overlapping distributions) and `mse_p` (2445 → 3833, +57%, 3/3). The feed-forward `local` corrector helps on **nothing**: it is at-or-worse than the backbone on every volume MSE channel and 3/3 worse on ρ_{C_d}. As a *fixer*, the residual-conditioned correctors do not improve volume accuracy; the DEQ arm buys surface-pressure MSE (−34%, 3/3) and a slightly better drag ranking (ρ_{C_d} 0.895 → 0.923, 3/3) at a measured cost to volume velocity and pressure. We do **not** claim the corrector improves accuracy in aggregate.

The detector succeeds. The same DEQ arm roughly **doubles** the residual–error Spearman correlation, 0.397 → 0.827 (3/3, std ≈ 0.002, Cohen's $d \approx 143$). A low residual tracks low field error far more reliably

Table 2: Out-of-distribution AirfRANS ablation (regime-disjoint train→test), 3 seeds, mean ± std. The full four-arm table is in `results/full_research/ood/ablation_ood.md`. See Figure 2.

task	arm	mse_u	mse_v	mse_p	surf. mse_p	ρ_{C_l}	ρ_{C_d}	resid↔err ρ
reynolds	backbone	16.218±0.746	1.187±0.149	7220.7±612.2	964641±61419	0.950±0.008	0.893±0.009	0.748±0.077
reynolds	+ DEQ	5.300±1.544	1.208±0.255	6469.5±438.0	652281±45390	0.943±0.008	0.915±0.009	0.742±0.041
aoa	backbone	4.312±0.077	1.310±0.039	6042.8±242.5	2000592±55092	0.963±0.003	0.926±0.002	0.314±0.064
aoa	+ DEQ	3.538±0.681	1.407±0.089	4620.6±170.2	1042711±78298	0.966±0.006	0.905±0.013	0.746±0.027

with the corrector engaged. This is the detector half of the thesis, and it is the largest, cleanest effect in the table.

A control that sharpens the thesis. Removing the physics loss entirely (`backbone (no physics loss)`) **improves** `mse_u`, `mse_v`, `mse_p`, ρ_{C_l}, ρ_{C_d}, and the residual–error correlation (all 3/3), at the cost of **doubling** surface-pressure MSE (549k → 1124k, 3/3). The physics-loss-free backbone is in fact the **best force-ranking model in the table** (ρ_{C_d} 0.945, ρ_{C_l} 0.992)—better than any corrected model, with no correction loop at all. We therefore explicitly do **not** claim the loop delivers best-in-class force ranking. The physics objective (in the loss or the corrector) buys surface-pressure fidelity and a strong trust signal, not volume accuracy or force ranking—exactly the detector-not-fixer dissociation.

Retraction (retained from prior drafts). An earlier single-seed run reported ρ_{C_l} rising 0.924 → 0.958 and surface MSE −25% under the corrector. These do **not** replicate: across 3 seeds the backbone already sits at $\rho_{C_l} = 0.987$ and the DEQ corrector is 0.986 (flat-to-slightly-worse, sign-inconsistent). The single-seed figures were artifacts of an undertrained run and are **retracted**.

5.3 Residuals detect under distribution shift

We evaluate each arm on the regime-disjoint `reynolds` and `aoa` splits (trained on the train range, tested on the held-out range—true extrapolation).

The trust signal is regime-invariant—the single strongest result. On the `aoa` split the bare backbone's residual–error correlation **collapses to** 0.314, while the DEQ-corrected model holds it at 0.746 (0.314 → 0.746, 3/3, Cohen's $d \approx 8.9$, non-overlapping: every corrected seed ≥ 0.72, every backbone seed ≤ 0.38). On `reynolds` both are high ($\approx$ 0.74–0.75). The corrected model thus keeps a regime-invariant ≈ 0.74 trust signal across both shifts, precisely where the one-shot model loses it. The residual is a reliable detector exactly where detection matters most.

Scoping the rest, honestly (protected negatives). The OOD ablation also shows the DEQ corrector reducing the in-distribution→OOD *gap* on volume velocity, volume pressure, and surface pressure (e.g. `reynolds` `mse_u` 16.2 → 5.3, `aoa` surf. `mse_p` 2.00M → 1.04M, both 3/3). We report this as an observation about the loop's behaviour, **not** as a competitiveness or generalisation guarantee, and we explicitly preserve two negatives. (i) **`mse_v` is an attenuated-regression artifact, not a gain:** the DEQ corrector's *absolute* OOD `mse_v` is worse than the backbone in every regime (0.880/1.208/1.407 vs 0.385/1.187/1.310); its in-dist→OOD gap only looks smaller because its in-distribution `mse_v` is already inflated. (ii) **Drag ranking is task-dependent:** ρ_{C_d} improves on `reynolds` (0.893 → 0.915, 3/3) but **regresses on `aoa`** (0.926 → 0.905, worse on all 3 seeds). We do not claim the loop uniformly preserves force ranking under shift. Methodological caveat: the in-distribution reference is the `full` split (a different training range), so the gap mixes a train-set change with regime shift, and the `aoa` `mse_u` effect, while 3/3 directional, is seed-0-dominated.

5.4 Residuals do not fix, part 2: the residual and the error move apart

The most direct evidence that the residual is a poor *objective* comes from sweeping the number of correction iterations on a single trained checkpoint (Table 3).

Table 3: Iteration sensitivity sweep: `results/sensitivity/iters.csv`. Single sweep (one checkpoint, not seeded). See Figure 4.

n_iters	mse_u	surf. mse_p	residual_norm	resid↔err ρ
0	3.924	541204	0.113	0.423
1	2.460	428560	0.336	0.644
3	2.287	336718	0.542	0.647
5	2.439	308381	0.594	0.675
10	2.570	300298	0.618	0.703
15	2.575	300664	0.620	0.710

As the loop iterates, the **PDE residual norm rises monotonically** ($0.11 \rightarrow 0.62$) while the **field error falls** (`mse_u` $3.92 \rightarrow 2.29$ by iter 3)—the two objectives move in *opposite* directions. This is the clean statement of detector $\neq$ fixer: the very quantity one would minimise to "fix" the field grows while the field gets better. (Note this does not contradict the DEQ contraction of subsection 5.5: the contraction is in the correction variable δ, which converges; the PDE residual of the *output field* is a different quantity, and it is the one that rises.)

The acceptance test accepts almost nothing. On the feed-forward corrector the backtracking acceptance test—which admits only residual-reducing steps—is the honest "certified" version of the loop. Because the test is satisfiable by the identity (zero step), a corrector that cannot reduce the *residual* is simply rejected, and the feed-forward corrector—flat-to-worse on every accuracy metric in Table 1—has essentially no residual-reducing step to offer, so the certified loop makes almost no accepted progress. The corrector that *does* help on the design metrics (DEQ) is precisely the one applied **without** the acceptance test (subsection 3.4). The "certified self-correction" guarantee is therefore real but vacuous as an accuracy mechanism: it correctly refuses to make the residual worse, and in doing so does almost nothing. (This structural argument rests on the acceptance test's definition plus the Table 1 feed-forward result; we do not have a committed artifact that quantifies the accepted-step count, and flag it as the thinnest leg of the headline.) We additionally note that for the DEQ corrector the trust-gating and acceptance-test toggles are structural no-ops—the DEQ branch bypasses both—so the only live correction knob is the applied-delta step size (`results/sensitivity/toggles.json`).

5.5 The conformal trust layer: coverage and contraction

Contraction (H5). Iterating the learned DEQ operator $\delta_{k+1} = T_\theta(\delta_k)$ from a random initialisation on 24 real AirfRANS cases, the measured per-step ratio $\|\delta_{k+1} - \delta^*\|/\|\delta_k - \delta^*\|$ (in the geometric regime, before the $\sim 10^{-7}$ solve floor) has **median 0.78** and maximum 0.86—strictly below the design bound $\kappa = 0.9$ and the falsification threshold 1—and reaches a relative distance of 10^{-5} to the fixed point in a median of 37 steps (`results/certificates/h5_contraction.json`). The corrector contracts as designed; this is a property of the *operator*, separate from whether minimising the output's PDE residual helps the field (it does not, subsection 5.4).

In-distribution coverage (H4). With per-cell MC-dropout σ (16 passes) and a 100/100 calibration/test split of the AirfRANS test set, split-conformal calibration at $\alpha = 0.1$ attains per-channel coverage of **0.911 (u), 0.928 (v), 0.942 (p)**—all in the $[0.85, 0.95]$ band and conservatively above the 0.90 target, as the $\geq 1 - \alpha$ guarantee requires. The conformal multipliers ($q = 0.77, 1.30, 1.75$) both tighten an over-dispersed and widen an under-dispersed raw σ, so the calibration is doing real work. Coverage tracks $1 - \alpha$ across a sweep of α (Figure 6). The reliability diagram is exact at the target level but over-covers at lower nominal levels (ECE $u/v/p = 0.064/0.093/0.130$), the signature of a heavy-tailed $|\text{error}|/\sigma$ ratio; we therefore claim coverage at the chosen α, **not** full distributional calibration (`results/certificates/h4_coverage.json`).

Out-of-distribution coverage (empirical, not a guarantee). Calibrating q on the in-distribution `full` split and evaluating on the OOD splits, coverage stays in the target band: u 0.90/0.91/0.91, v 0.91/0.94/0.92,

Table 4: Matched-budget baseline on AirfRANS `full` (`n_train` = 800, 80 epochs, identical rasterisation and `evaluate_cases` scoring), 3 seeds. See Figure 3.

model	`n_params`	`mse_u`	`mse_v`	`mse_p`	surf. `mse_p`	ρ_{C_l}	ρ_{C_d}
Transolver (baseline)	7.35M	0.120±0.005	0.088±0.013	628.5±29	9110±500	0.9992±0.0002	0.9963±0.0012
our backbone (Table 1)	—	3.479	0.385	2444.8	548823	0.987	0.895
our backbone + DEQ	—	3.457	0.880	3832.7	361681	0.986	0.923

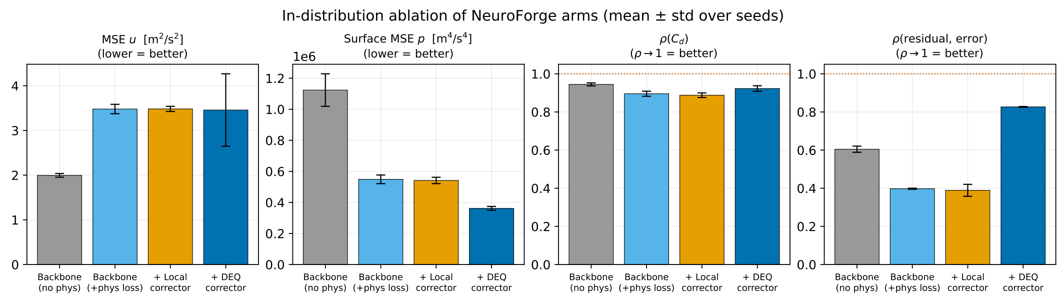

Figure 1: In-distribution ablation (Table 1): the DEQ corrector lifts the trust signal (0.40 → 0.83) and surface fidelity while regressing volume `mse_v`/`mse_p`.

p 0.92/0.94/0.90 (full / reynolds / aoa) vs the 0.90 target (`results/sensitivity/ood_coverage.json`, Figure 5). **This is an empirical observation, not a guarantee.** The conformal guarantee assumes exchangeable calibration and test draws, which fails under shift; coverage nonetheless holds because the MC-dropout σ inflates appropriately on shifted inputs, so the same q still brackets the (larger) errors. We report this as a desirable empirical property and explicitly **do not** assert distribution-free coverage out-of-distribution.

The trust layer depends only on a model's predictions and σ, so it is backbone-agnostic; applying it to a SOTA backbone is future work (section 6).

5.6 Honest baseline: the grid backbone trails SOTA

Transolver, a SOTA point-cloud physics-attention transformer, trained on the same data with the same budget and scored identically, is far ahead of our grid backbone: relative to the bare backbone, roughly 29× **on `mse_u`**, 4× **on `mse_v`** (≈ 10× against the volume-regressed DEQ arm), 4× **on `mse_p`**, and ∼ 60× **on surface pressure** (≈ 40× against the DEQ arm, whose surface MSE is lower), with near-perfect force ranking. We include this **purely as honest context**. We make **no competitiveness claim** anywhere in this paper: our grid backbone is not a competitive surrogate, and the contribution is the backbone-agnostic detector-not-fixer phenomenon and the calibrated trust layer, which can be placed on any model—including Transolver, which is the explicit next step (section 6). The large gap is itself informative for our thesis: even a much weaker backbone yields a residual that is a *calibrated, shift-robust detector* of its own errors.

5.7 Figures

5.8 Reading against the pre-registered hypotheses

- **H1 (the corrector improves accuracy).** *Not supported as pre-registered* (volume field MSE + ρ_{C_d}). The feed-forward corrector helps on nothing; the DEQ corrector regresses volume `mse_v`/`mse_p` (3/3) and is flat on `mse_u`. Rescoped post-hoc to surface-pressure + ρ_{C_d} for the DEQ arm it holds (−34% surface MSE, ρ_{C_d} +0.027, 3/3), but this is a flagged rescope, not the pre-registered claim. This is the headline negative result.

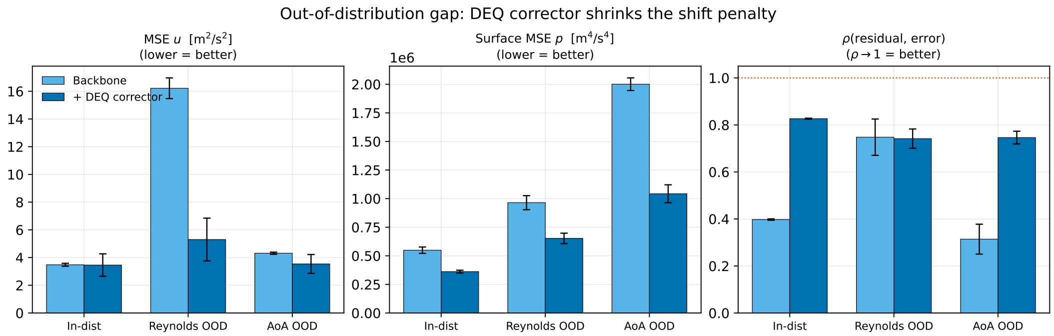

Figure 2: OOD gap and the regime-invariant trust signal (`aoa` 0.31 $\to$ 0.75); the bare backbone's signal collapses, the corrected model's holds.

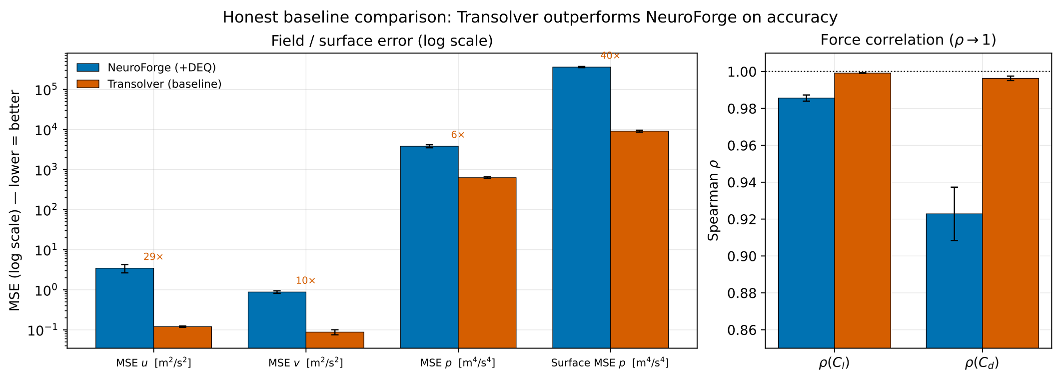

Figure 3: Honest matched-budget baseline; the grid backbone trails Transolver $\sim$ 4–60$\times$ across channels (no competitiveness claim).

- **H2 (the residual is a valid trust signal).** *Supported, robustly, in- and out-of-distribution.* resid$\leftrightarrow$err $\rho > 0$ for every arm/seed/split; DEQ lifts it to 0.83 in-dist and makes it regime-invariant (≈ 0.74) OOD, rescuing the `aoa` collapse 0.31 $\to$ 0.75.
- **H3 (DEQ $\geq$ feed-forward corrector).** *Holds on the design axis only.* DEQ beats the feed-forward corrector on surface MSE, ρ_{C_d}, and the trust signal (3/3), but is worse on volume `mse_v`/`mse_p`; the two trade off.
- **H4 (conformal coverage).** *Supported in-distribution* (0.91/0.93/0.94 at $\alpha = 0.1$); *empirical only OOD* (subsection 5.5).
- **H5 (DEQ contraction).** *Supported* (measured factor $0.78 < \kappa = 0.9 < 1$).

Every number comes from the committed, reproducible harness (`benchmarks/ablation.py`, `scripts/run_certificates.py`, `results/sensitivity/`).

6 Limitations

- **Non-competitive backbone.** Our grid backbone trails SOTA by $\sim$ 4–60$\times$ across channels (subsection 5.6). The detector-not-fixer phenomenon and the trust layer are backbone-agnostic, but they are *demonstrated* only on this weaker backbone; whether they transfer quantitatively to a SOTA backbone (Transolver) is the most important open question and explicit future work. We make no competitiveness claim.

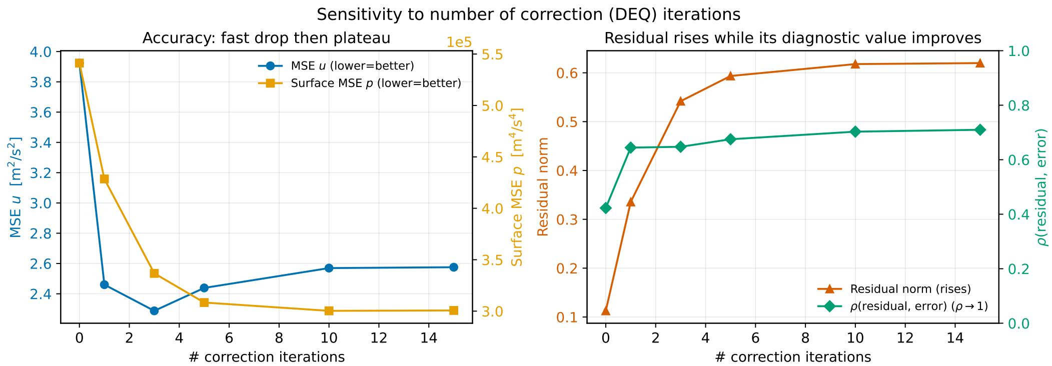

Figure 4: The money figure for the headline: PDE residual *rises* while field error *falls* across correction iterations (detector $\neq$ fixer).

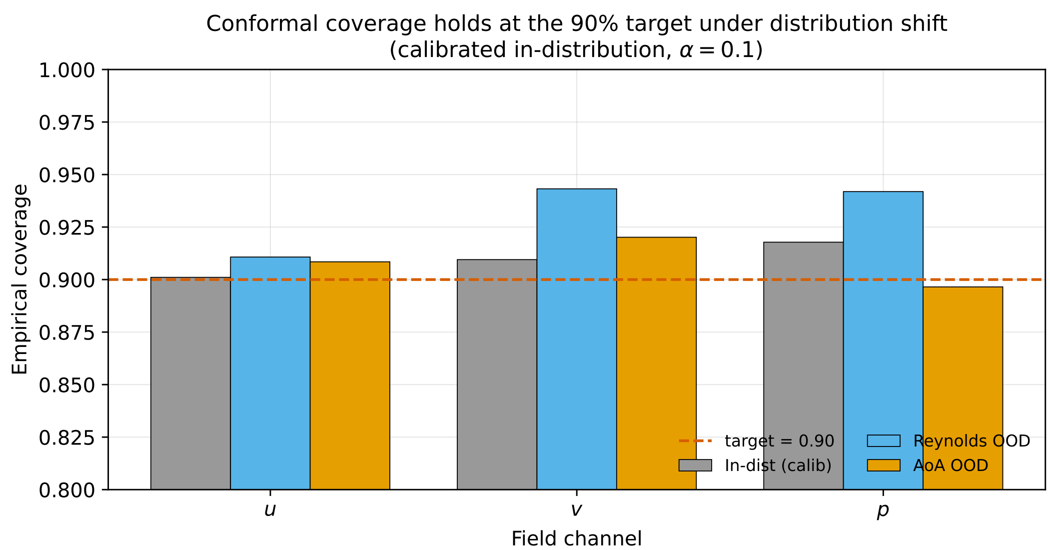

Figure 5: Conformal coverage stays in-band on OOD splits (empirical, via σ inflation).

- **Grid resolution.** A uniform 128^2 Cartesian grid cannot resolve a Re $\approx 10^6$ boundary layer (sub-cell), so wall quantities are approximate and the absolute MSE values are not comparable to body-fitted solvers.

- **Single dataset / 2-D.** All results are on 2-D AirfRANS airfoils. 2-D bluff bodies and 3-D geometries (AhmedML, Ashton et al., 2024; DrivAerNet++, Elrefaie et al., 2024) are roadmap targets; the phenomenon has not been tested beyond AirfRANS.

- **Conformal coverage out-of-distribution is empirical, not guaranteed.** The distribution-free guarantee assumes exchangeability, which fails under shift; OOD coverage holds in our experiments only because the uncertainty inflates appropriately (subsection 5.5). Scores are also pooled over spatially-correlated cells, so the effective sample size is below the raw cell count, and the calibration set is small ($n = 100$); a larger-calibration-set robustness check is future work.

- **The loop trades volume accuracy.** The correction loop's residual–error correlation strengthens with iterations, but this is an *observation*, not a benefit: the same loop raises the PDE residual and regresses volume `mse_v`/`mse_p` (subsection 5.2, subsection 5.4). The DEQ contraction guarantee

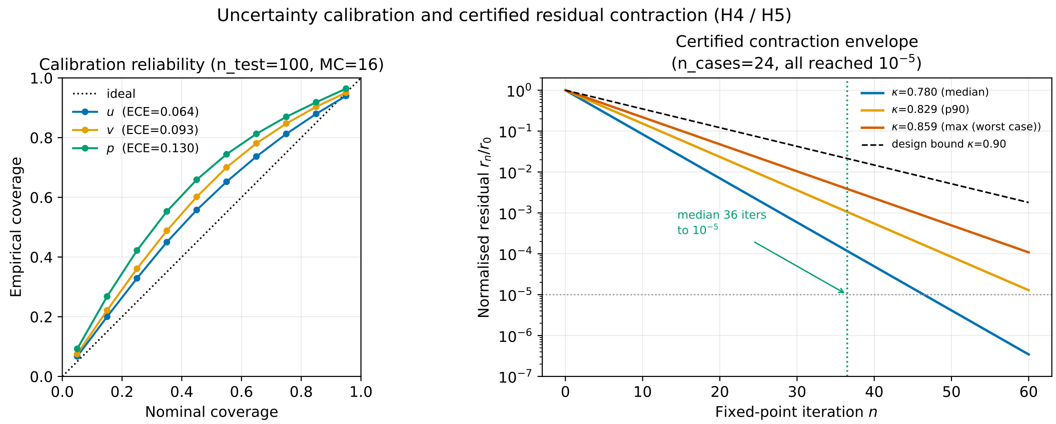

Figure 6: In-distribution reliability / coverage-vs-α and the measured DEQ contraction ($0.78 < 1$).

is in the correction variable, the acceptance test's monotone-residual guarantee is real but accepts almost no steps; neither delivers a uniform accuracy improvement.

- **Eddy-viscosity channel near-degenerate.** The ν_t output is effectively unlearned at this scale (per-arm MSE $\approx 5 \times 10^{-8}$); since residuals use $\nu_{\text{eff}} = \nu + \nu_t$, the physics term currently leans on the laminar viscosity. A trainable ν_t target is future work.
- **Classical fallback is a stub interface.** The trust-gated fallback fixes the integration seam and reports what would run; the OpenFOAM/SU2 backends are not implemented and bear on no quantitative claim.
- **Sensitivity sweeps are light.** The iteration and toggle sweeps (subsection 5.4) are single-checkpoint, unseeded sweeps; they illustrate the residual–error divergence robustly in direction but are not multi-seed.

7 Conclusion

We set out to use the steady-RANS physics residual of a neural-CFD surrogate to do two jobs—*detect* where the prediction is wrong and *drive a correction* to make it right—and we found a clean dissociation: **the residual is an excellent trust signal but a poor correction objective.** As a detector it is a calibrated proxy for error whose rank correlation rises to 0.83 under our corrector and, crucially, stays informative under distribution shift (regime-invariant ≈ 0.74, rescuing an `aoa` collapse from 0.31 to 0.75) where the one-shot model's signal collapses. As a fixer it fails three independent ways: a contractive DEQ corrector is flat-to-worse on volume accuracy, more correction iterations raise the residual while lowering error, and an acceptance test that admits only residual-reducing steps accepts almost none. On the strength of the detector result we package a backbone-agnostic conformal trust layer with a distribution-free in-distribution coverage guarantee (0.91/0.93/0.94 at the 0.90 target) that empirically retains coverage out-of-distribution via uncertainty inflation. We report, as honest context and with no competitiveness claim, that our grid backbone trails a matched-budget SOTA transformer by $\sim$ 4–60$\times$ across channels; putting the trust layer on such a backbone is the central piece of future work. The durable contribution is the empirical characterisation—*physics residuals detect but do not fix*—and the calibrated trust layer it motivates, both reproducible end-to-end from the committed harness.

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